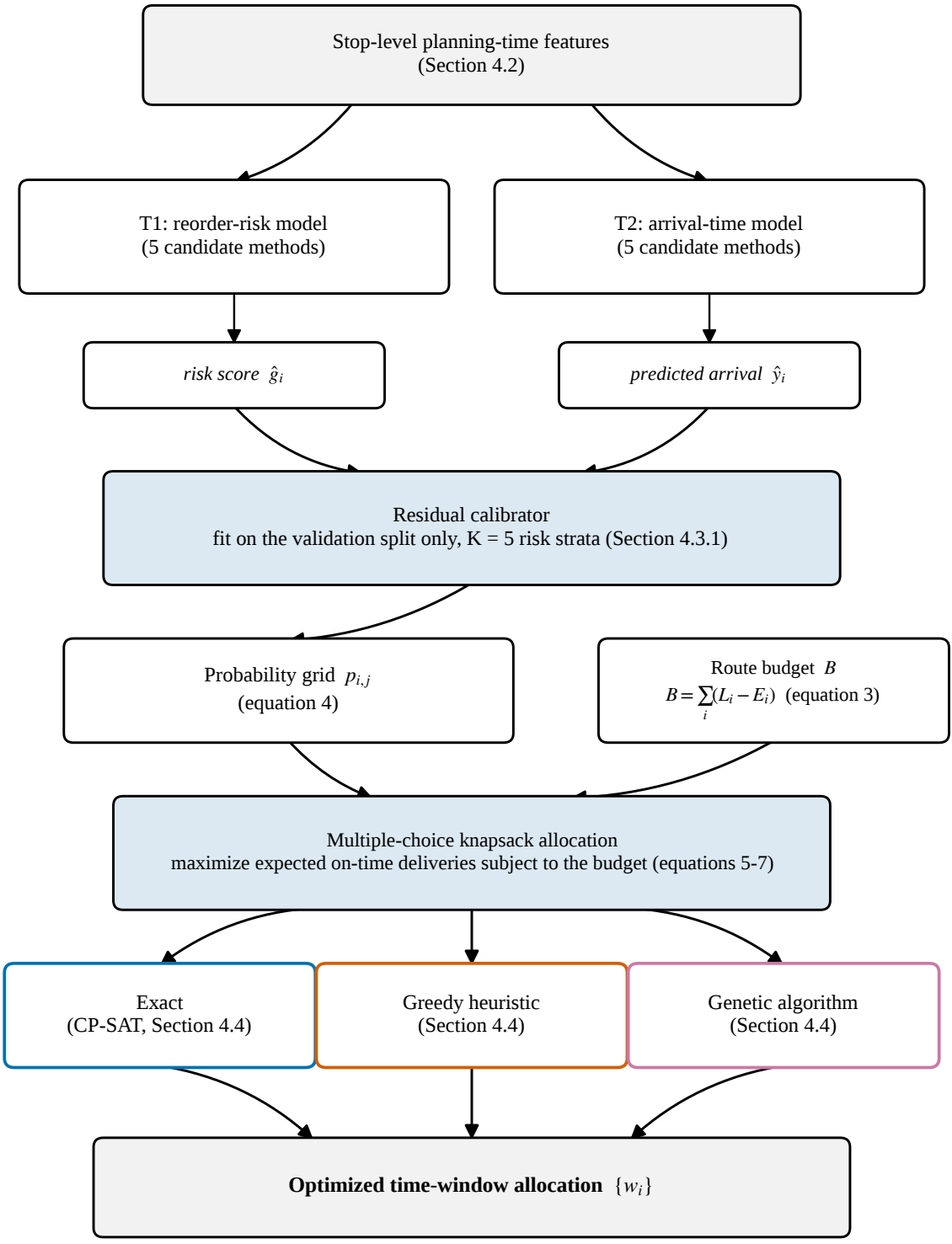
Stop-level planning-time features
(Section 4.2)
T1: reorder-risk model
(5 candidate methods)
T2: arrival-time model
(5 candidate methods)
risk score $\hat{g}_i$
predicted arrival $\hat{y}_i$
Residual calibrator
fit on the validation split only, K = 5 risk strata (Section 4.3.1)
Probability grid $p_{i,j}$
(equation 4)
Route budget B
$B = \sum_i (L_i - E_i)$ (equation 3)
Multiple-choice knapsack allocation
maximize expected on-time deliveries subject to the budget (equations 5-7)
Exact
(CP-SAT, Section 4.4)
Greedy heuristic
(Section 4.4)
Genetic algorithm
(Section 4.4)
Optimized time-window allocation $\{w_i\}$